maddy1112-exe Update README.md 2979fa8 · now

Arithmetic-operation-using-8086

8086 Assembly Language Programs for Arithmetic Operations

AIM

To write and execute Assembly Language Programs to perform arithmetic operations for the 8086 microprocessor.

APPARATUS REQUIRED

- Personal Computer with MASM Software

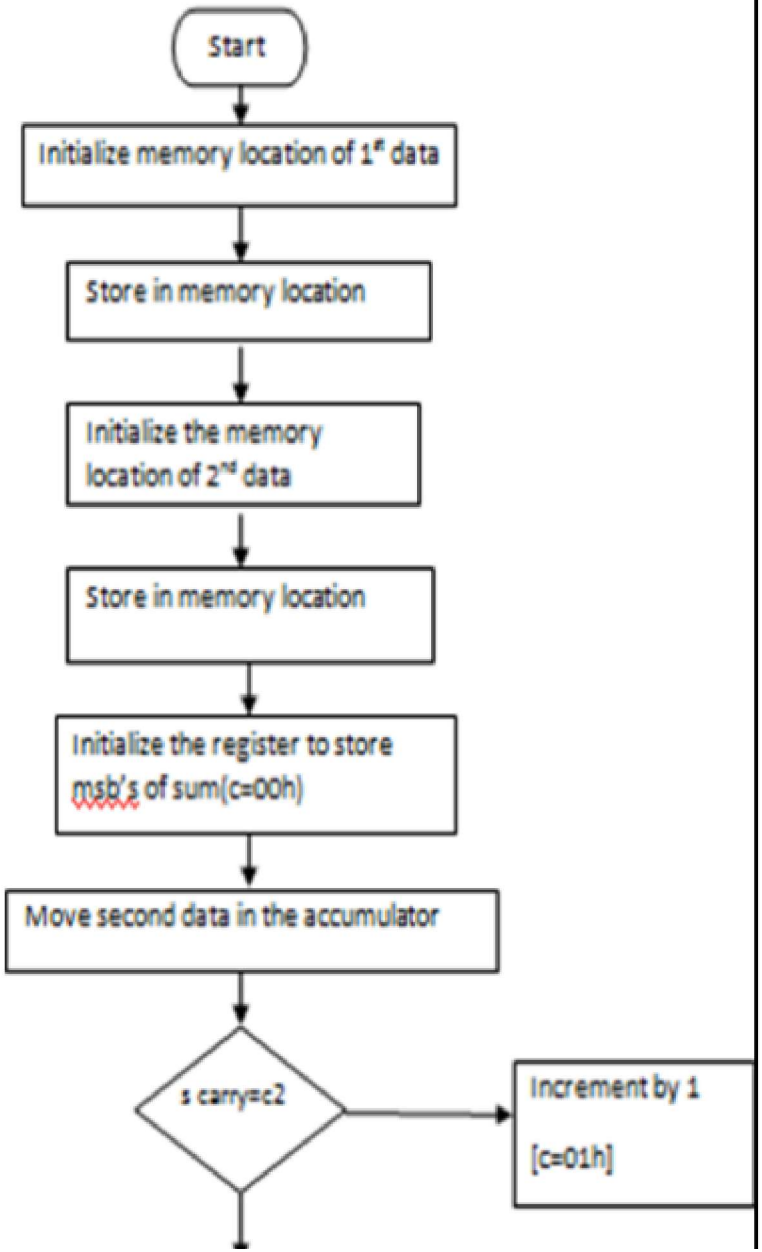
1. ADDITION

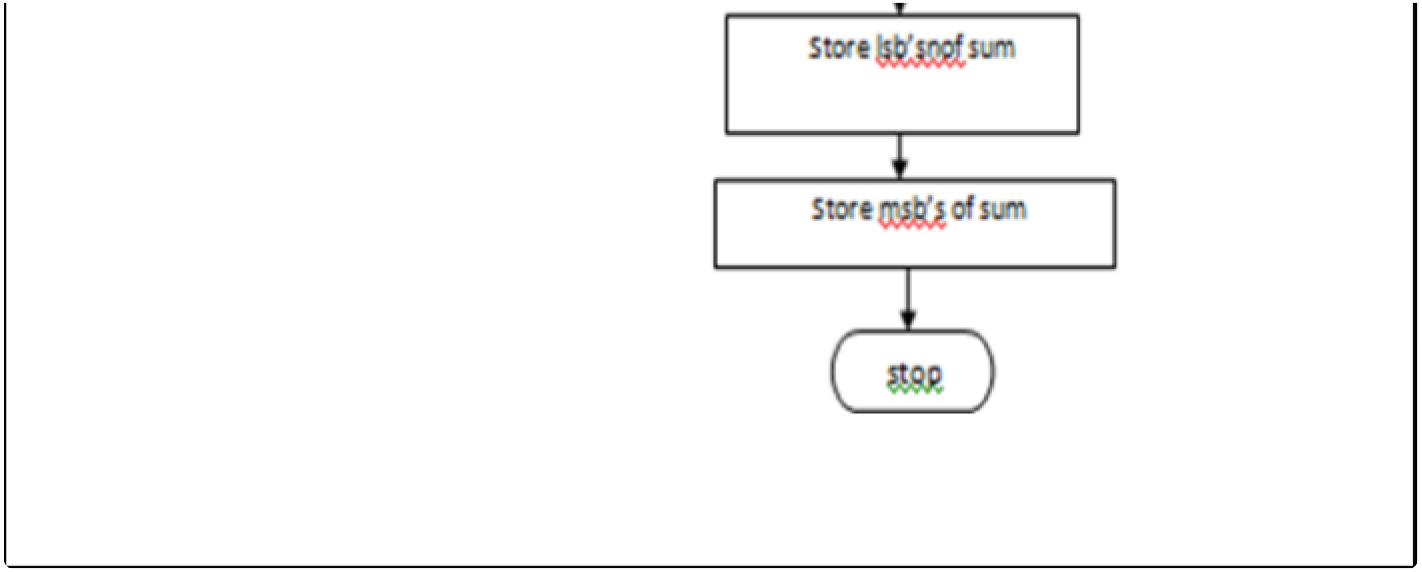
Algorithm

1. Initialize memory location in HL register.
2. Store 1st data.
3. Increment HL to enter 2nd data.
4. Move 2nd number to accumulator.
5. Decrement HL.
6. Add value in memory with accumulator.
7. Store result.
8. Stop.

FLOW CHART

FLOW CHART:(indirect)





Program

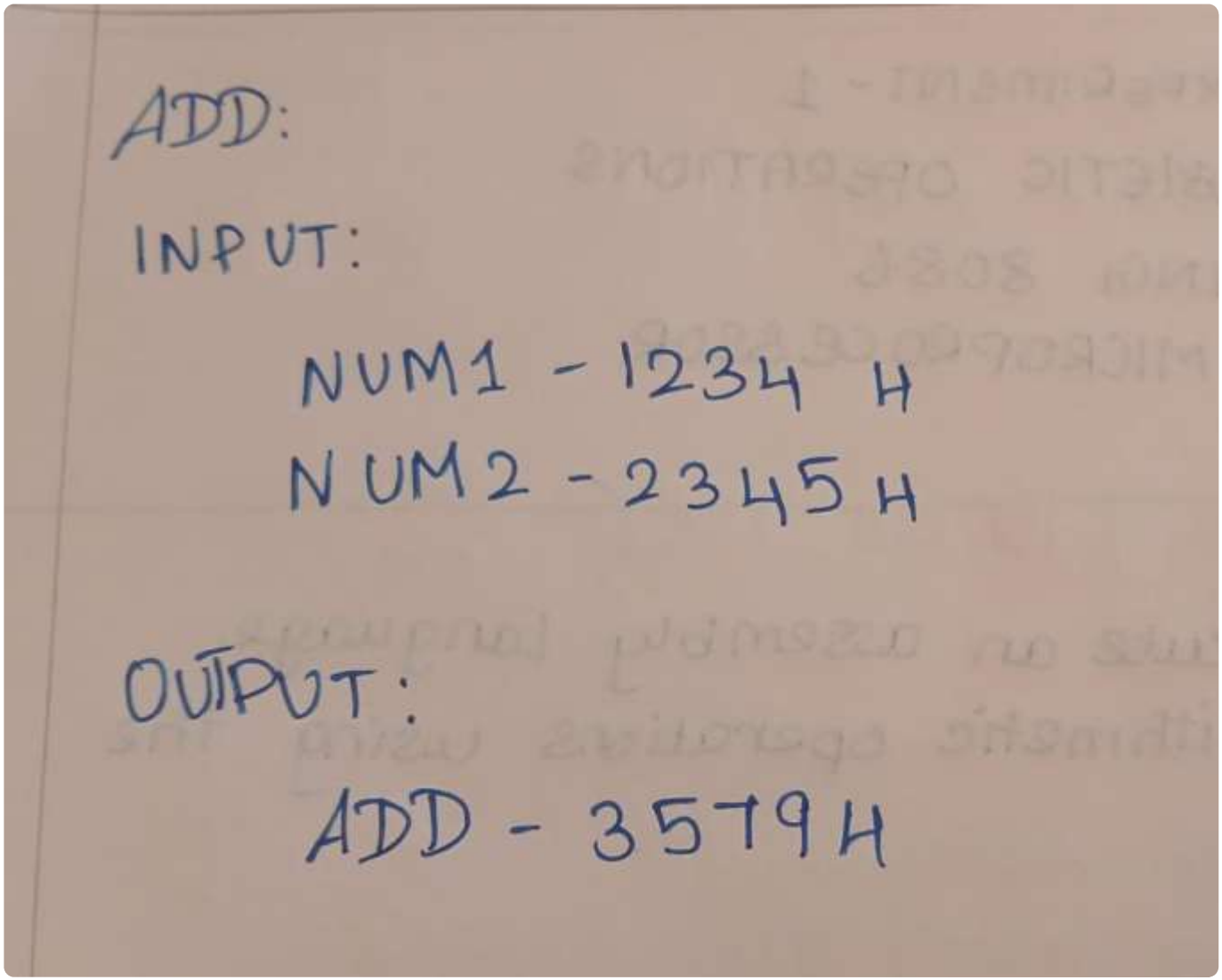
```
CODE SEGMENT
ASSUME CS:CODE, DS:CODE
ORG 1000H
MOV CL,00H
MOV AX,1234H
MOV BX,1234H
ADD AX,BX
JNC L1
INC CL
L1:MOV SI,1200H
MOV [SI],AX
MOV [SI+2],CL
MOV AH,4CH
INT 21H
CODE ENDS
END
```

Output Table

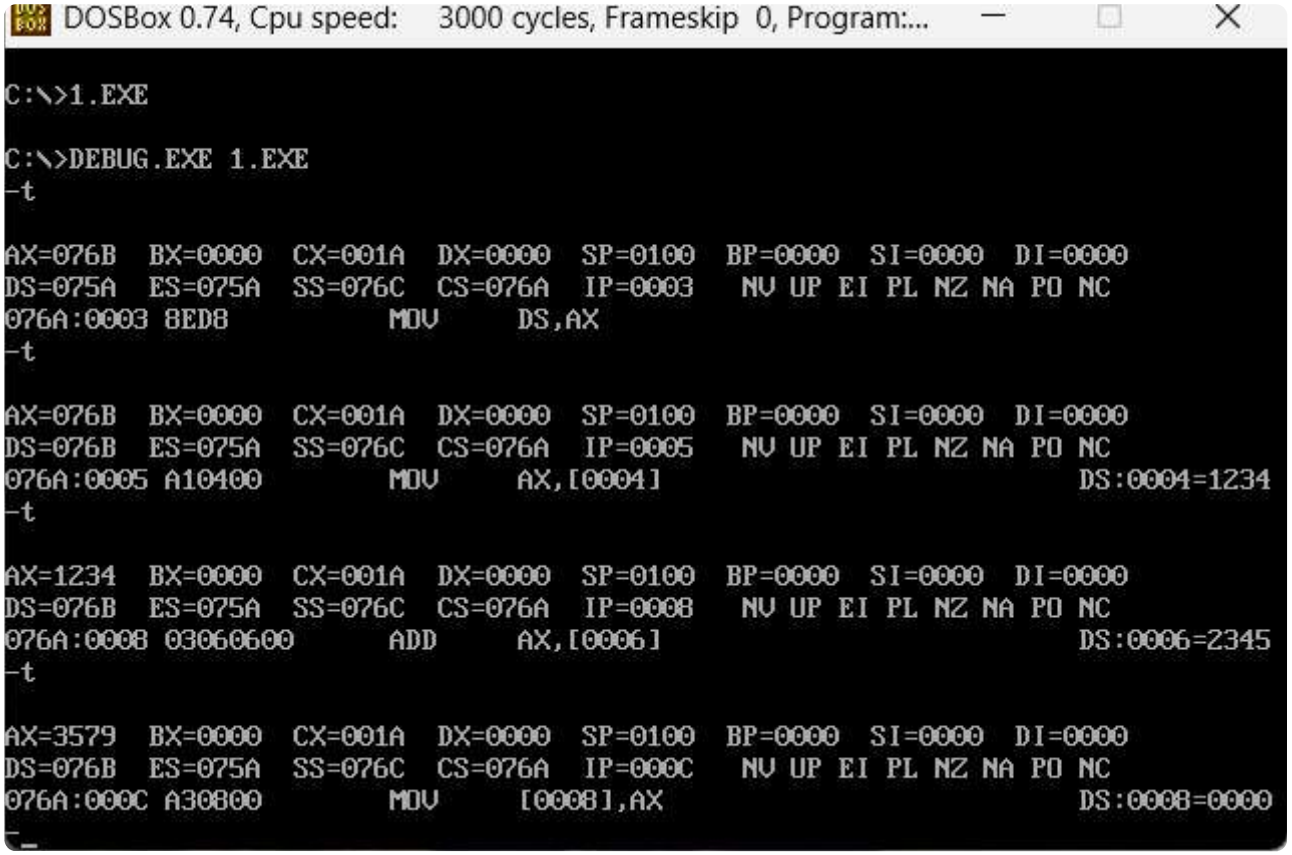
MEMORY LOCATION (INPUT)	MEMORY LOCATION (OUTPUT)
1200 🔢 01 12	

| 1200 |

Manual Calculations



OUTPUT IMAGE FROM MASM SOFTWARE



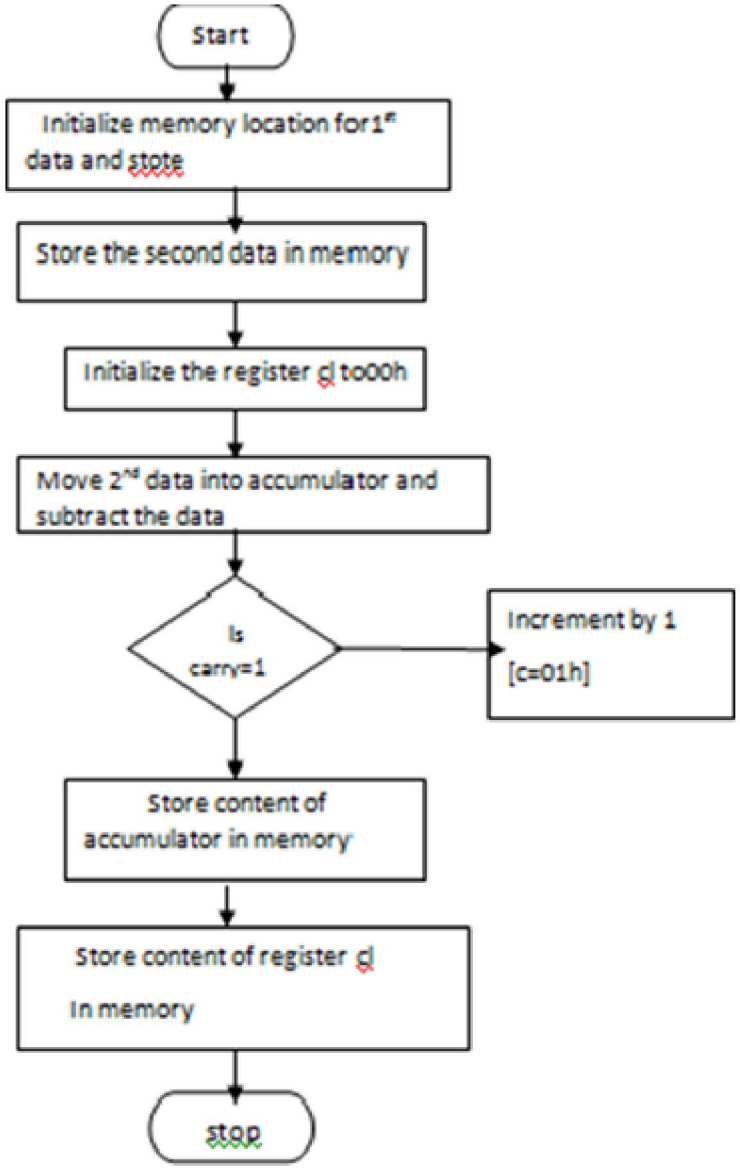
2. SUBTRACTION

Algorithm

1. Initialize memory and store 1st data.
2. Increment to get 2nd data.
3. Move 2nd data to accumulator.
4. Subtract memory content.

5. Store result.

FLOWCHART


Start
Initialize memory location for1st data and stote
Store the second data in memory
Initialize the register cl to00h
Move 2nd data into accumulator and subtract the data
Is carry=1
Increment by 1 [c=01h]
Store content of accumulator in memory
Store content of register cl In memory
stop

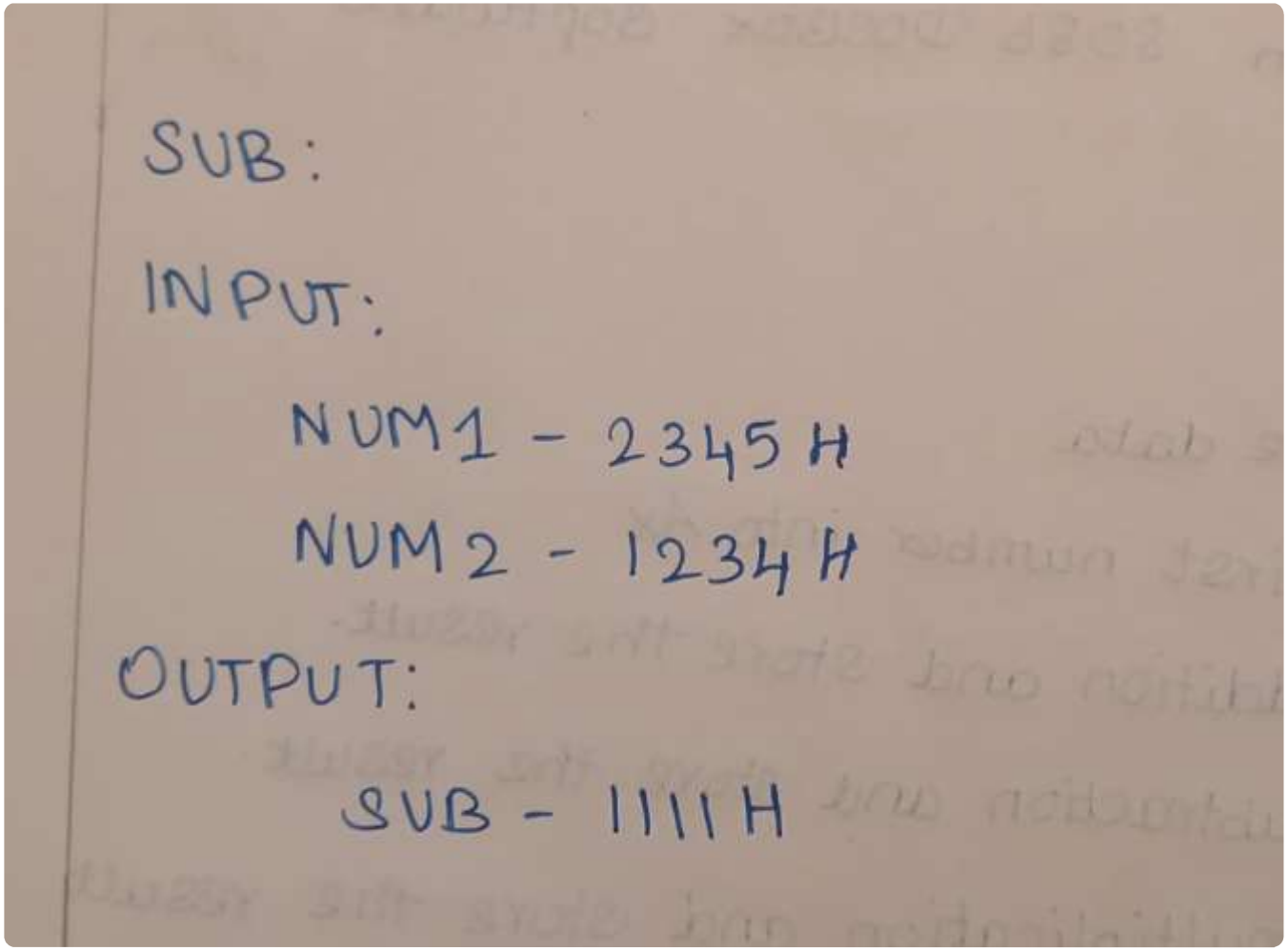
Program

```
CODE SEGMENT
ASSUME CS: CODE, DS: CODE
ORG 1000H
MOV SI,2000H
MOV CL,00H
MOV AX,[SI]
MOV BX,[SI+02H]
SUB AX,BX
JNC L1
INC CL
L1:
MOV [SI+04H],AX
MOV [SI+06H],CL
MOV AH,4CH
INT 21H
CODE ENDS
END
```

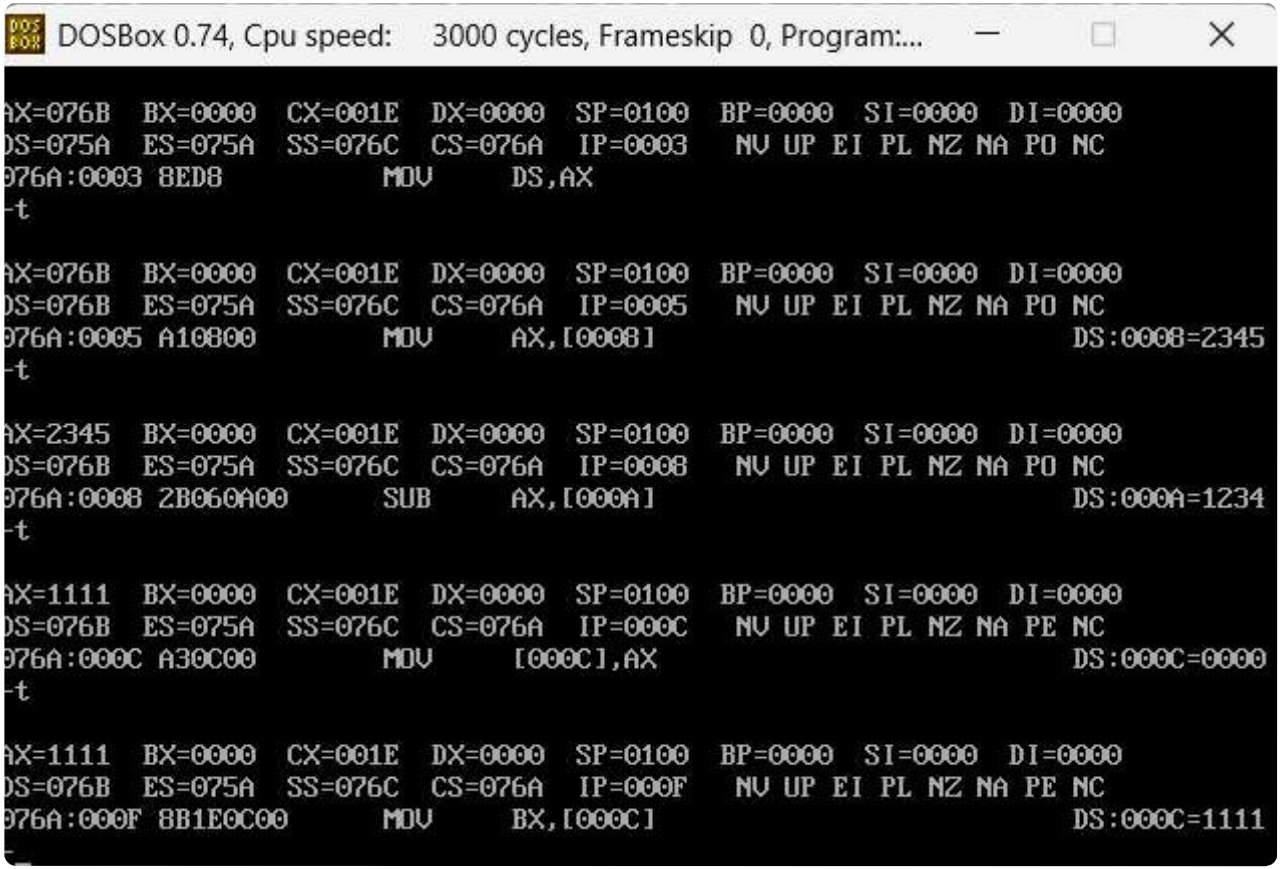
Output Table

MEMORY LOCATION (INPUT)	MEMORY LOCATION (OUTPUT)

Manual Calculations



OUTPUT SCREEN FROM MASM SOFTWARE

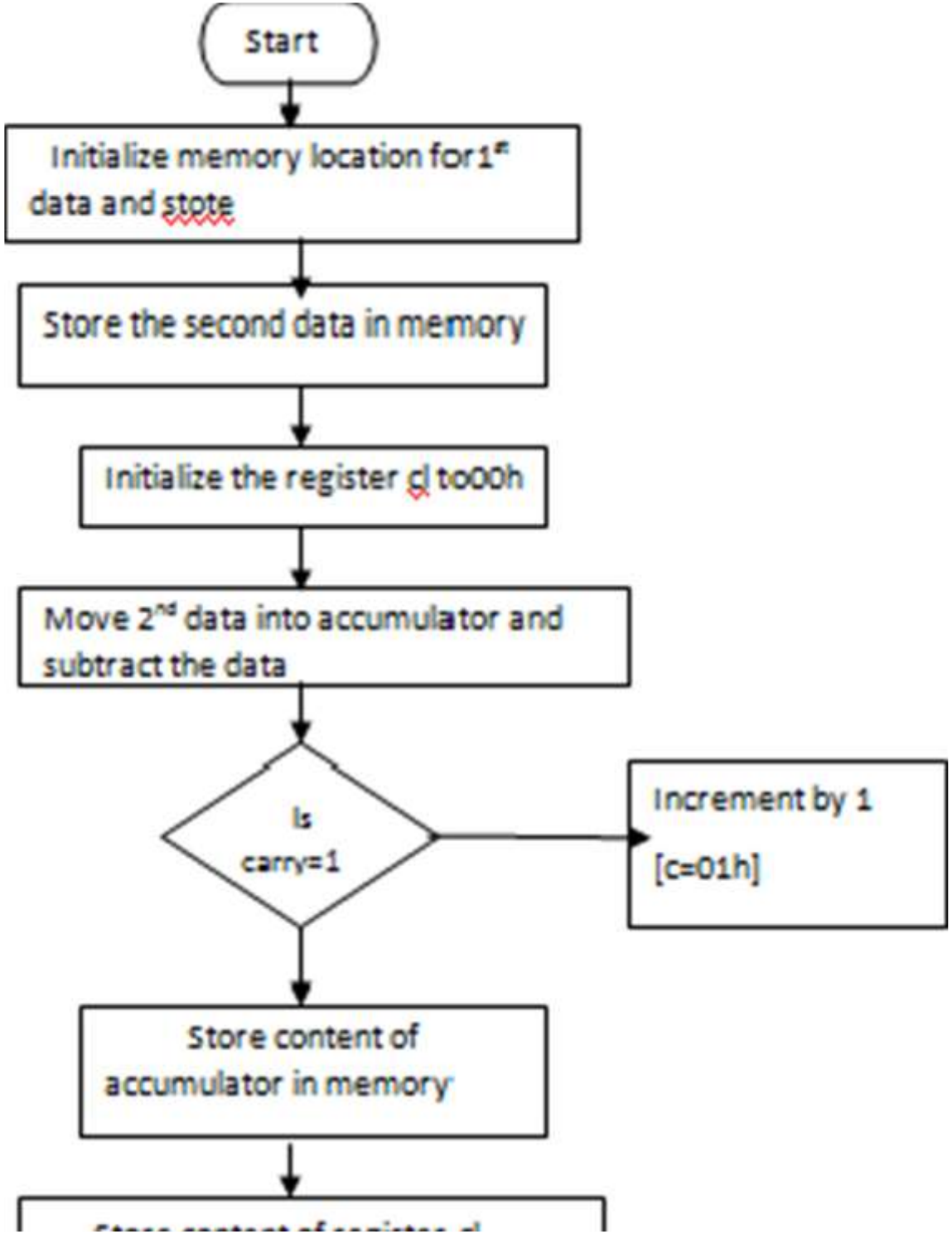


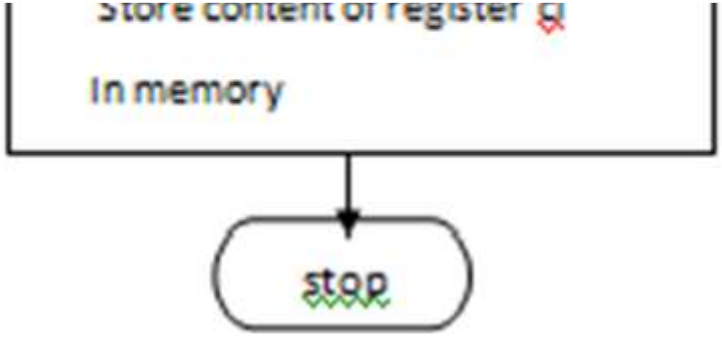
3. MULTIPLICATION

Algorithm

1. Initialize memory and store operands.
2. Move operands to registers.
3. Multiply.
4. Store result.

##FLOWCHART


Start
Initialize memory location for 1st data and stote
Store the second data in memory
Initialize the register cl to00h
Move 2nd data into accumulator and subtract the data
Is carry=1
Increment by 1 [c=01h]
Store content of accumulator in memory



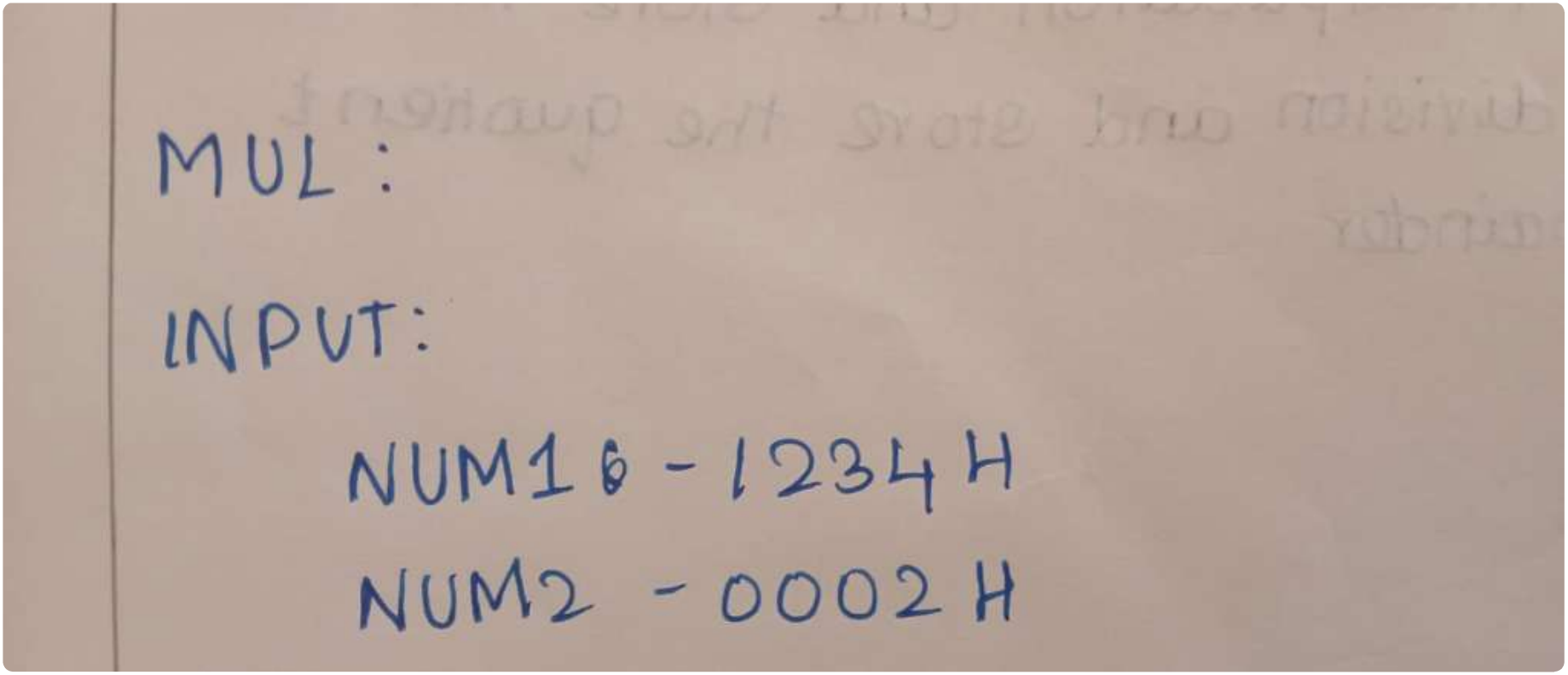
Program

```
CODE SEGMENT
ASSUME CS: CODE, DS: CODE
ORG 1000H
MOV SI,2000H
MOV DX,0000H
MOV AX,[SI]
MOV BX,[SI+02H]
MUL BX
MOV [SI+04H],AX
MOV [SI+06H],DX
MOV AH,4CH
INT 21H
CODE ENDS
END
```

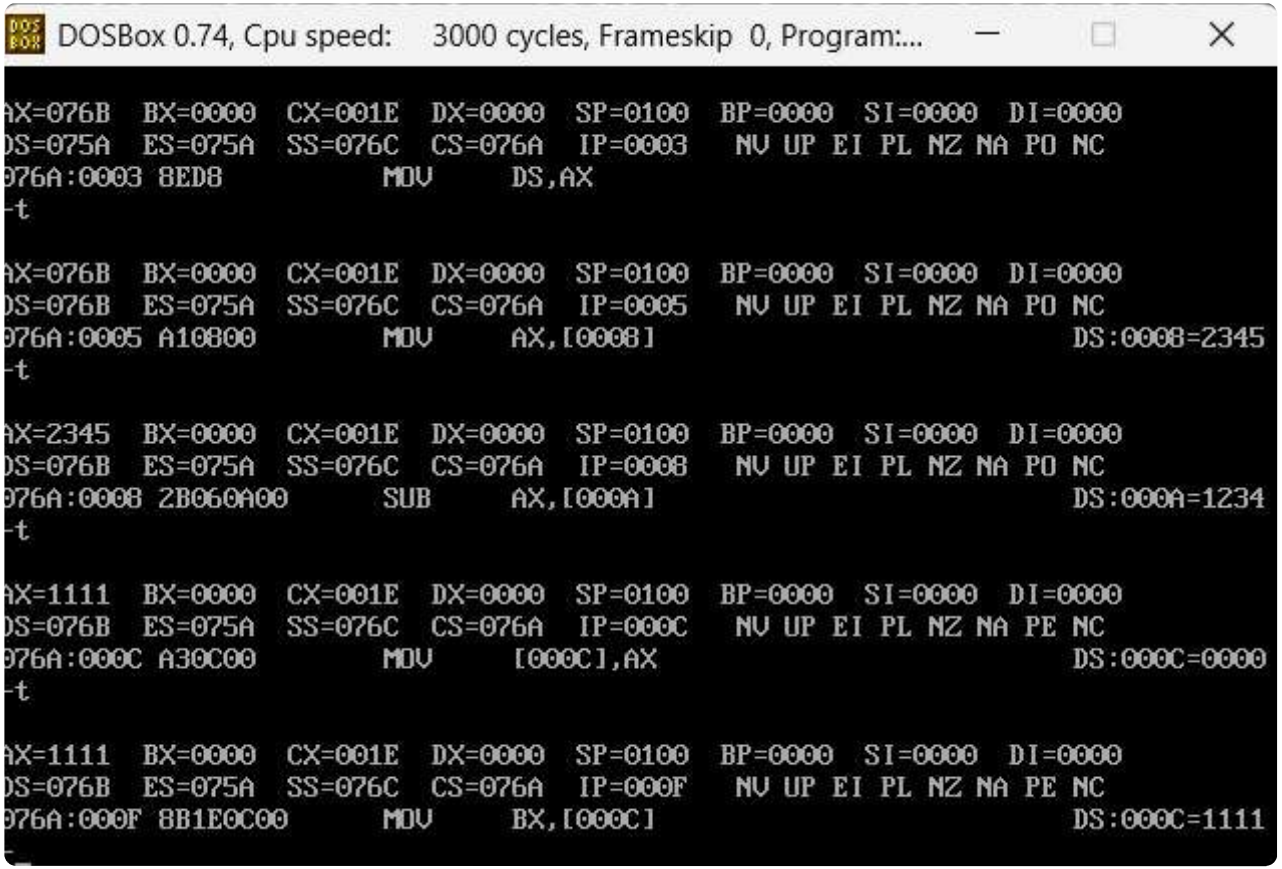
Output Table

MEMORY LOCATION (INPUT)	MEMORY LOCATION (OUTPUT)

Manual Calculations



OUTPUT SCREEN FROM MASM SOFTWARE



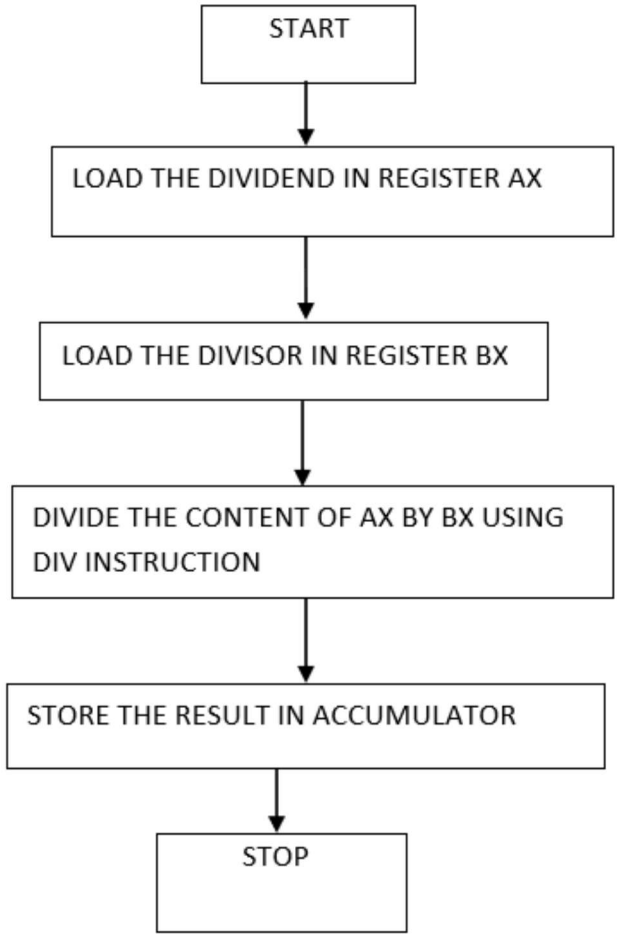
4. DIVISION

Algorithm

1. Load memory location of operands.
2. Perform division.
3. Store result.

FLOWCHART

Flowchart (INDIRECT METHOD):



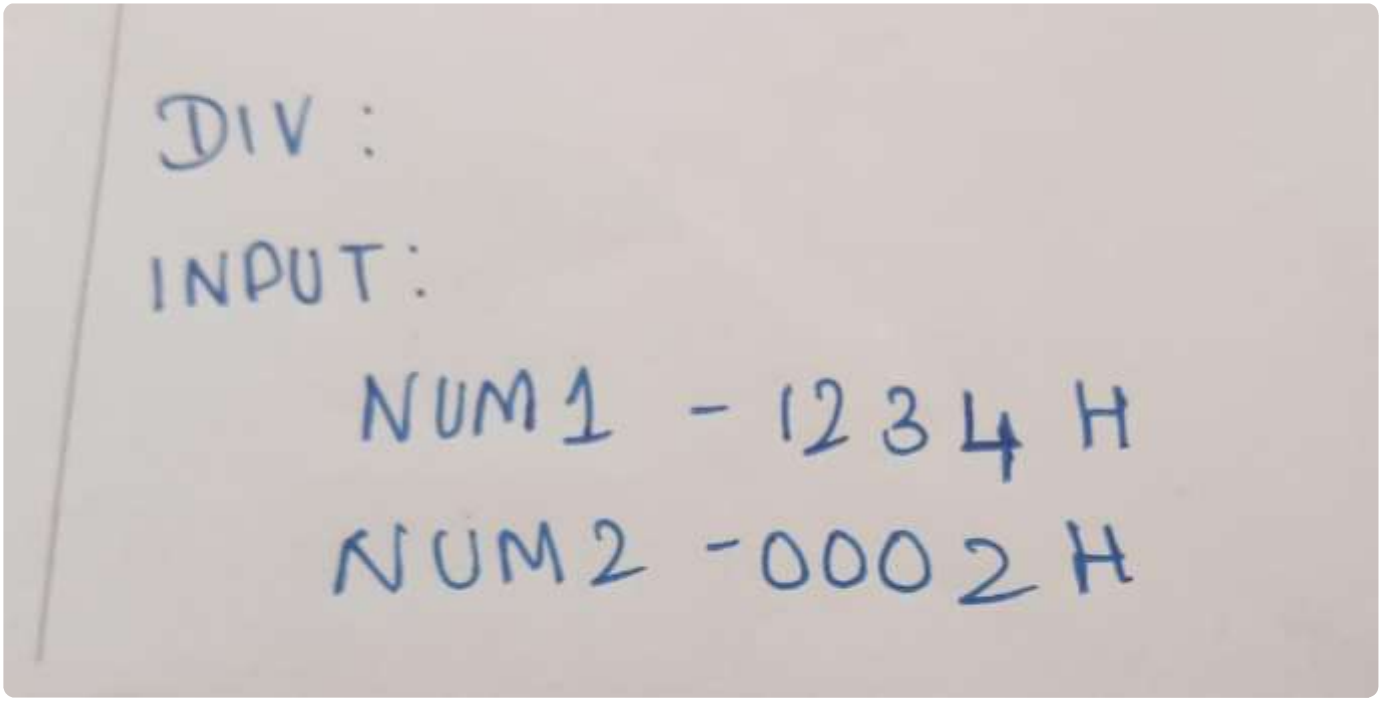
Program

```
CODE SEGMENT
ASSUME CS: CODE, DS: CODE
ORG 1000H
MOV SI,2000H
MOV DX,0000H
MOV AX,[SI]
MOV BX,[SI+02H]
DIV BX
MOV [SI+04H],AX
MOV [SI+06H],DX
MOV AH,4CH
INT 21H
CODE ENDS
END
```

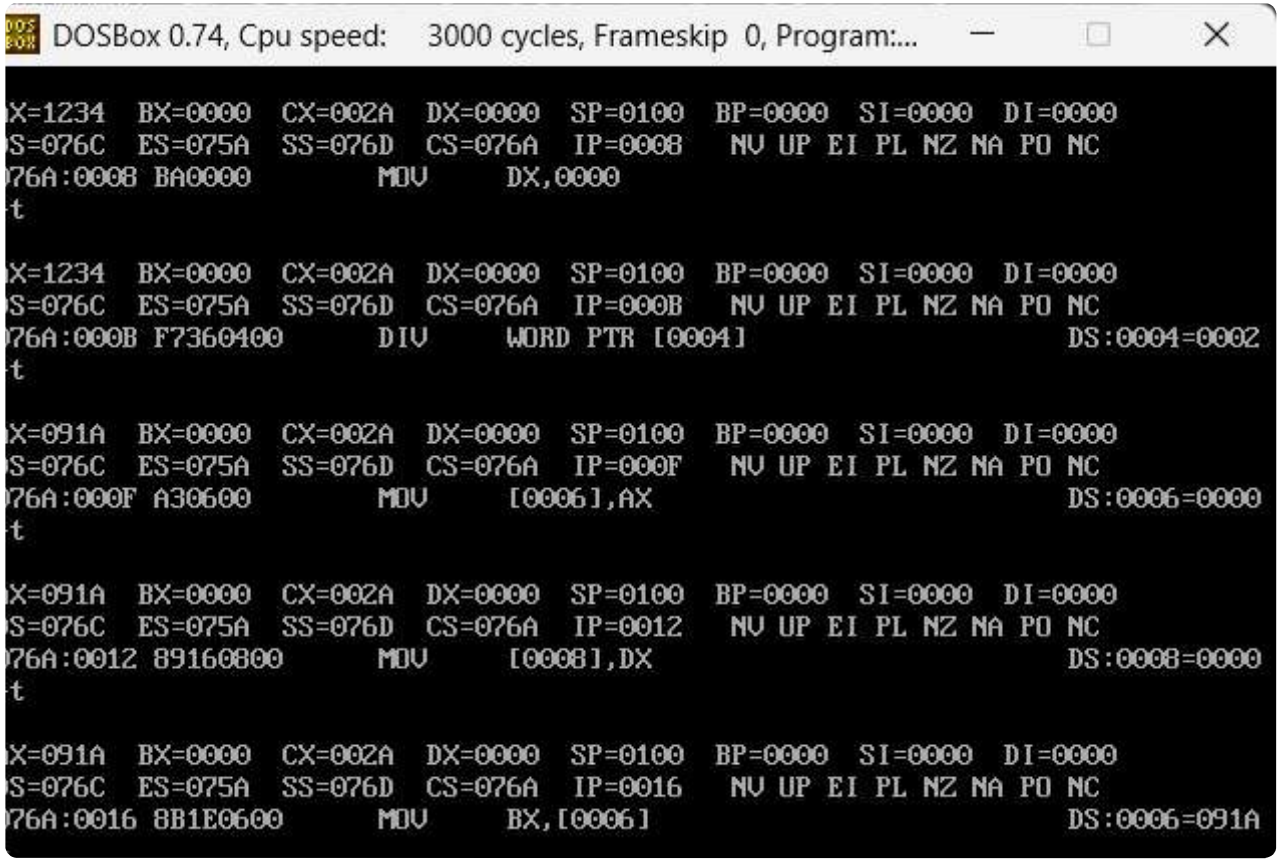
Output Table

MEMORY LOCATION (INPUT)	MEMORY LOCATION (OUTPUT)

Manual Calculations



OUTPUT FROM MASM SOFTWARE



RESULT

Thus, the Assembly Language Programs for 8086 to perform arithmetic operations (Addition, Subtraction, Multiplication, and Division) using both direct and indirect methods were successfully written and executed using MASM.